Introduction and Scope

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Preface

The Statistical Architecture Framework (SAF) was developed to support the modernisation of official statistics. This framework responds to the need of national statistical institutes and their partners for shared guidelines and standards. Producing statistics must remain credible, secure, and efficient. At the same time, organisations must adapt quickly to new data sources, higher user expectations, and tighter regulations and budgets. The SAF provides a solid foundation for this and help organisations guide these changes in a controlled and structured way.

This new framework replaces the older CSPA and CSDA standards, but it reuses the elements that are still current and valuable. The SAF combines the core ideas from these previous framework into a single, logical, and coherent model. While the framework continues to focus on cooperation, reuse, and standards, the focus has shifted heavily toward a practical application of the framework in an organisation. It is specifically designed to be used directly within ongoing programmes and projects. This helps teams make concrete choices when designing and implementing new processes and solutions.

The fundamental principle of the framework is that data and its descriptions are the most important assets of a statistical organisation. Information is not just a tool to run processes, but the core of the organisation. The SAF therefore treats this information as an NSIs most valuable asset that must be professionally managed throughout its entire lifecycle. It must support systems that can work with both traditional and modern, new data sources and information flows. In addition, it offers a practical guide for collecting, combining, and publishing data.

The SAF builds on a modular approach where systems are constructed from loose, flexible components. It promotes the reuse of these digital building blocks, both within an organisation and between different organisations. This prevents projects from independently building the same tools, which reduces duplication and lowers costs. Furthermore, it ensures a clear separation between business methods, data, and the

underlying technology. This separation is essential for keeping complex IT landscapes manageable and successfully scaling them up. The framework is designed to be used at multiple levels at the same time. It helps with strategic planning, designing specific IT solutions, and monitoring how different projects align with one another.

The SAF does not replace existing statistical models, but rather works closely alongside them. It forms the connecting link between organisational strategy, business processes, and technology. This connection helps organisations make consistent decisions and makes sure that organisation policy is translated into practice. To improve communication between teams, the SAF uses different visual models. By using a standard notation style, organisations can easily compare and reuse each other's designs. Finally, the SAF is a living framework that will continuously evolve based on practical experience, community feedback, and new technological developments.

Acknowledgements

TODO

Introduction

The Statistical Architecture Framework (SAF) provides a common architectural basis for the design, delivery, and evolution of statistical production in National Statistical Institutes (NSIs) and related organisations. It brings together architectural concepts, reference structures, and practical guidance that support consistent decision-making across portfolios and programmes. SAF aims to help statistical organisations modernise with control, while preserving the institutional qualities that make official statistics credible: transparency, quality, impartiality, reproducibility, and public trust.

Statistical organisations are facing a period of accelerated change. Data ecosystems are evolving and increasingly depend on partnerships and shared infrastructures. Artificial intelligence and rapid technological advances are changing how data can be collected, processed, and interpreted, while also introducing new risks and new expectations. At the same time, trust, scientific rigor, and data protection requirements are becoming more prominent and more demanding. These changes occur in an uncertain and fragmented global environment, where information integrity and societal confidence cannot be taken for granted. All of this is happening under sustained resource constraints, requiring organisations to be more agile and deliberate in how they prioritise, reuse, and industrialise capabilities. SAF is intended to support organisations in quickly responding to these drivers without fragmenting their architecture, governance, and implementation landscape in the process.

SAF replaces earlier reference documentation by consolidating core architectural ideas into a single framework that is more usable in practice. It is not designed as a theoretical reference alone. It is designed to be applied. SAF supports the translation of strategy into capabilities, services, and implementation choices. It supports alignment across business, methodology, data, applications, and technology. It also supports collaboration between organisations by promoting common patterns, shared viewpoints, and comparable architectural descriptions.

Purpose

The purpose of SAF is to provide an authoritative, consistent, and actionable framework for developing and governing statistical architectures. It aims to support coherent modernisation across NSIs by providing shared structures for describing needs, designing solutions, and managing change. SAF helps ensure that investments in platforms, data management, methods, and delivery practices strengthen each other instead of competing. It also helps organisations to demonstrate traceability from policy and strategy to implemented solutions, including the controls required for confidentiality, integrity, accountability, and lawful processing.

SAF is intended to enable better outcomes in three areas. First, interoperability and reuse, both within an organisation and across organisational boundaries.

Second, quality and robustness, by making architectural choices explicit and reviewable, and by promoting consistent handling of data, metadata, and methodological assets. Third, overall statistical production effectiveness, by providing a common language and set of expectations that connects architects, methodologists, engineers, managers, and oversight functions.

Definition of Scope

SAF covers the enterprise architecture concerns that are most relevant to statistical production and dissemination, including the supporting functions that enable them. It addresses the design and evolution of capabilities, information assets, applications and services, and underlying infrastructure and deployment arrangements. It also considers the cross-cutting concerns that are essential for official statistics, such as governance, security, privacy, auditability, reproducibility and the responsible use of data and advanced analytics.

To ensure consistency and practical applicability, SAF is organised according to a TOGAF-aligned structure. It addresses motivation and principles, strategic direction and target outcomes, and the architecture domains that connect intent to implementation. This includes business architecture, data architecture, application architecture, and technology and infrastructure considerations, as well as deployment and operationalisation aspects. SAF also includes viewpoints that are tailored to distinct roles and activities within an NSI, enabling the same underlying architecture to be communicated and assessed from multiple perspectives without losing coherence.

SAF does not prescribe a single operating model, a single technology stack, or a single organisational design. It recognises that NSIs differ in mandate, maturity, and context. Instead, it provides a shared architectural foundation that can be adapted to local realities while remaining comparable across organisations. It also assumes that terminology and detailed definitions are managed through a dedicated glossary, which is maintained alongside the framework.

Target Audience

This documentation is primarily written for enterprise and data architects, IT architects, methodologists, and IT delivery teams. It is also intended for senior decision-makers such as CIOs and CTOs, and for privacy officers, policy makers, and managers who carry responsibility for compliance, traceability, and responsible data use.

These groups and individuals can use this framework as a reference and as a working instrument. It can be applied when shaping strategy and roadmaps, when designing new capabilities and platforms, and when assessing initiatives for alignment and risk. It can be used to structure architectural work in programmes and projects, and to support governance decisions by making assumptions, dependencies, and trade-offs explicit. It can also be used to support communication across disciplines by establishing shared architectural concepts and consistent viewpoints.

Different roles will use SAF in different ways, while relying on the same underlying content. Enterprise and data architects will use it to define target states, standards, patterns, and architecture decisions across the organisation. IT architects and delivery teams will use it to translate those decisions into solution designs, implementation plans, and operational controls. Methodologists will use it to connect methods and process design to the enabling data and technology arrangements, and to support repeatability and scientific rigor. CIOs, CTOs, managers, and policy makers will use it to understand the implications of strategic choices, to prioritise investment, and to oversee risk and compliance. Privacy officers and assurance functions will use it to assess whether designs incorporate adequate safeguards, traceability, and accountability.

SAF is most effective when it is used consistently across the lifecycle of change. It supports early-stage framing by clarifying drivers, objectives, and constraints. It supports design by providing reference structures and viewpoints that guide architecture development. It supports delivery by improving alignment between architecture and implementation decisions. It supports operation and evolution by making dependencies and responsibilities explicit, enabling controlled change and measurable improvement.

Expected value

The expected value of SAF is a stronger and more coherent statistical architecture practice across NSIs. It supports faster and safer modernisation by reducing ambiguity and rework. It supports reuse by encouraging common building blocks and comparable solution descriptions. It supports trust by embedding governance and assurance concerns into architecture, rather than treating them as afterthoughts. It also supports collaboration across organisations, where shared approaches to interoperability, data governance, and service orientation reduce the effort required to work together. In providing a common framework with practical orientation, SAF aims to help statistical organisations remain resilient and responsive. It supports them in delivering high-quality statistics under changing conditions, while maintaining the safeguards and institutional discipline that official statistics require.

Constraints & Assumptions

Use of TOGAF and ArchiMate in SAF

The SAF builds on two open standards of The Open Group, a global, vendor-neutral consortium that develops technology and architecture standards: the TOGAF® Standard and the ArchiMate® Specification. This section explains what these standards are, why the SAF uses them, and how they relate to each other and to the statistical standards of the UNECE community.

What TOGAF is

The TOGAF Standard is a framework for developing and governing enterprise architecture. It provides:

- **A method.** The Architecture Development Method (ADM) is an iterative cycle of phases. It runs from establishing the architecture capability and an architecture vision, through business, information systems, and technology architecture, to migration planning, implementation governance, and change management.
- **A content framework.** A common structure for the artefacts that architecture work produces, such as principles, stakeholder maps, capability models, and architecture building blocks.
- **Guidance for an architecture capability.** Advice on governance, roles, and maturity, so that architecture becomes a lasting practice rather than a one-off project.

The SAF refers to the TOGAF Standard, 10th Edition. More information is available at The Open Group: TOGAF Standard.

Why the SAF uses TOGAF

The SAF uses TOGAF for three reasons.

- **A shared structure:** Many NSIs already use TOGAF, or a framework derived from it, in their own architecture practice. Aligning the SAF with TOGAF lets NSIs map SAF content onto their existing repositories, governance processes, and training without translation.
- **A proven method:** TOGAF describes how to move from motivation to implementation in controlled steps. The SAF follows the same order: motivation, strategy, business architecture, data, application, and technology architecture, and implementation. This gives NSIs a recognisable path from intent to solution.
- **A common language across organisations:** TOGAF terminology is widely understood by architects, IT suppliers, and international partners. Using it reduces ambiguity when NSIs compare architectures or collaborate on shared solutions.

The SAF uses TOGAF as a **structuring reference, not as a prescription**. It tailors the TOGAF structure to official statistics and does not require NSIs to follow every ADM phase or to hold TOGAF certification. The SAF does not reproduce TOGAF content; readers are referred to the standard for its full description.

What ArchiMate is and how it relates to TOGAF

ArchiMate is a modelling language for enterprise architecture. Where TOGAF describes *how* to develop an architecture and *which* content to produce, ArchiMate describes *how to express* that content in consistent, unambiguous models and views. The two standards are maintained by The Open Group and are designed to work together:

Aspect	TOGAF	ArchiMate
Nature	Framework and method	Modelling language and notation
Answers	How is architecture developed and governed?	How is architecture described and visualised?
Main content	ADM phases, content framework, architecture capability	Elements, relationships, viewpoints
Role in the SAF	Structures the chapters and the order of the framework	Notation for the models and diagrams in the SAF

The structures of the two standards correspond. The TOGAF Preliminary Phase, Phase A, and Requirements Management correspond to the ArchiMate motivation and strategy elements. TOGAF Phases B, C, and D correspond to the ArchiMate business, application, and technology layers. TOGAF Phases E to H correspond to the ArchiMate implementation and migration elements. This correspondence is the basis for the position of each SAF chapter. The figure groups the ADM phases in three areas:

- **Strategy and Motivation:** the Preliminary Phase, Phase A (Architecture Vision), and Requirements Management.
- **Core layers:** Phases B, C, and D, which produce the Business, Application (Information Systems), and Technology architectures.
- **Implementation and Migration:** Phases E to H, which plan, govern, and change the realisation of the architecture.

PLACEHOLDER: Logical model of the different layers to be added

More information is available at The Open Group: ArchiMate.

Relationship with statistical standards

TOGAF and ArchiMate are generic standards. The SAF complements them with the standards of the statistical community: GSBPM for statistical business processes, GAMSO for the activities of statistical organisations, GSIM for statistical information objects, and SDMX for the exchange of statistical data and metadata. TOGAF and ArchiMate provide the architectural structure and notation; the statistical standards provide the domain content. The SAF connects the two.

Assumptions and constraints

ID	Type	Statement
SAF-A1	Assumption	The SAF is structured according to the TOGAF Standard, 10th Edition, tailored to official statistics.
SAF-A2	Assumption	Models and diagrams in the SAF use the ArchiMate 3.2 Specification. Alignment with the ArchiMate 4 Specification will be assessed in a later version of the SAF.

ID	Type	Statement
SAF-A3	Assumption	NSIs can apply the SAF without TOGAF or ArchiMate certification. Basic familiarity with both standards helps but is not required.
SAF-A4	Assumption	Terminology and definitions are maintained in a separate SAF glossary, which takes precedence over definitions in this document.
SAF-C1	Constraint	The SAF refers to TOGAF and ArchiMate but does not reproduce their content. Use of the standards is subject to the licence terms of The Open Group.
SAF-C2	Constraint	The SAF does not prescribe a technology stack, tool, or organisational design (see *Definition of Scope*).

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Reading guide

This reading guide explains how the SAF is built, in which order to read it, and how an NSI can start using it. It also explains how the SAF fits into an NSI's own architecture, how it combines with other reference models, and how legislation enters the architecture.

1. How the SAF is structured

The SAF follows the layers of TOGAF and ArchiMate (see *Constraints & Assumptions*). Each layer answers one question. Each layer is justified by the layer above it, and each layer gives direction to the layer below it.

PLACEHOLDER: Overview of the SAF layers and chapters

Chapter	Question it answers	Main content	Status [**RG4**]
Motivation	*Why* does the NSI change, and what does it want to achieve?	Stakeholders, drivers, assessments, goals, outcomes, principles, requirements, constraints, meaning, value	Developed
Strategy	*What* must the NSI be able to do, and how does it get there?	Capability model, value streams, resources, courses of action and strategies	Developed

Chapter	Question it answers	Main content	Status [**RG4**]
Business Architecture	*How* is the statistical work organised?	Business domains, reference models for the data value chain (Steady-State, Medallion), steady states, value stream detail, roles	In development
Data Architecture	*Which* data and metadata are needed, and how are they structured?	To be developed	Planned
Application Architecture	*Which* applications and services support the work?	To be developed	Planned
Technology Architecture	*On which* platforms and infrastructure does it run?	To be developed	Planned
Implementation and Tooling	*How* is the architecture realised and governed?	To be developed	Planned

Every section that describes an architecture concept follows the same pattern:

1. **The concept.** What the concept means, based on ArchiMate and TOGAF.
2. **How the SAF applies it.** Why the SAF uses the concept, and how it fits into an NSI's own architecture.
3. **Relationships with other concepts.** How the concept connects to the rest of the framework.
4. **SAF elements or examples.** The reference content of the SAF, or illustrative examples where the content is specific to each NSI.
5. **Toolkit.** A reference to templates and instruments in the SAF toolkit.

This pattern makes it possible to read a single section without reading the whole document, and still see how it connects to the rest.

The SAF contains two kinds of content. **Reference content** is shared by most NSIs and can be adopted directly, such as the drivers, goals, principles, capability model and value streams. **Illustrative content** shows the form of elements that each NSI defines for itself, such as assessments, requirements, constraints, resources and courses of action. Illustrative content is never a requirement.

2. Reading order

The SAF is best read top-down, at least once: from Motivation, through Strategy, to the architecture domains. This order follows the logic of the framework. A business architecture choice only makes sense once the goals and capabilities behind it are clear.

Applying the SAF is different from reading it. Application is iterative. An NSI does not complete the Motivation layer before it starts on its business architecture. It works in cycles, like the TOGAF Architecture Development Method: each cycle refines the motivation, strategy and architecture a little further, and each change in a driver or goal can start a new cycle.

Different readers can follow different routes through the document (see *Target Audience* for the roles).

Reader	Start with	Then read	Focus
CIO, CTO, board member	Introduction, Motivation	Strategy: capability model, courses of action	Why change is needed, what it delivers, how to prioritise

Reader	Start with	Then read	Focus
Enterprise or data architect	Introduction, Motivation	Strategy, Business Architecture, later domains	The full chain, and how to adopt it in the NSI architecture
IT architect, delivery team	Motivation: principles	Business Architecture, later domains	Design rules and reference models for solutions
Methodologist	Strategy: value streams	Business Architecture: steady states	Where methods are applied, and how quality and lineage are secured
Privacy officer, auditor	Motivation: principles, constraints	Business Architecture: steady states; VS-07	Where protection and compliance are built into the chain
Policy maker, manager	Introduction, Motivation: goals, value	Strategy: courses of action	What the framework means for the organisation

3. Positioning the SAF in your own architecture

TOGAF describes a range of architectures, from generic to specific, called the *Enterprise Continuum* [**RG2**]. It runs from foundation architectures, which apply to any organisation, through common systems and industry architectures, to the architecture of one specific organisation.

Level in the Enterprise Continuum	Example	Who owns it
Foundation architecture	TOGAF, ArchiMate	The Open Group
Common systems architecture	Generic reference architectures for, for example, data platforms or identity management	Industry and standards bodies
Industry architecture	**The SAF**, for official statistics	UNECE statistical community
Organisation-specific architecture	The enterprise architecture of an NSI	The NSI

The SAF is an industry architecture. It is not the architecture of any single NSI. Each NSI builds its own organisation-specific architecture, using the SAF as its reference. In practice, this works as follows:

1. **Adopt, adapt or extend.** For each SAF element, the NSI decides whether to adopt it as it is, adapt it to the national context, or add NSI-specific elements:
   - **Adopt.** Use an SAF element as is, when it matches the national situation.
   - **Adapt.** Refine an SAF element with national specifics, such as a national statistics act or a government-wide digital strategy.
   - **Extend.** Add elements that are specific to the NSI, while keeping a reference to the closest SAF element where one exists.

   The decisions are recorded in SAF-TK-G02 Adoption register.
2. **Keep the SAF identifiers.** When an NSI adapts an SAF element, it keeps a reference to the SAF identifier, for example SAF-P07 or capability 2.1.2. This keeps the NSI architecture comparable with the SAF and with other NSIs, and makes it easier to adopt future SAF versions.
3. **Layer the architecture.** The NSI enterprise architecture sets the direction for solution architectures in programmes and projects. Solution architectures do not refer to the SAF directly, but to the NSI architecture that applies it.

PLACEHOLDER: From SAF reference architecture to NSI enterprise architecture to solution architecture

4. Combining the SAF with other reference models

The SAF rarely stands alone. NSIs also work with statistical standards, with reference models of their own government, and with generic frameworks. These models complement each other, but they can also overlap or conflict.

Family	Examples	Role next to the SAF
Statistical standards and models	GSBPM, GAMSO, GSIM, SDMX; the ESS Enterprise Architecture Reference Framework (ESS EARF)	Provide the statistical domain content. The SAF connects them in one architecture
Government reference models	The European Interoperability Framework (EIF) and Reference Architecture (EIRA); national reference models such as those listed below	Often mandatory or leading for public organisations. The SAF fits within them
Generic frameworks and bodies of knowledge	TOGAF, ArchiMate, DAMA-DMBOK	Provide structure, notation and professional practice

Examples of national government reference models [**RG3**]:

Country	Reference model	Owner
Netherlands	NORA, Netherlands Government Reference Architecture	Dutch government (ICTU)
United Kingdom	GovS 005 Digital, Government Functional Standard	Cabinet Office, Government Digital and Data
Canada	Government of Canada Enterprise Architecture Framework	Treasury Board of Canada Secretariat
Australia	Australian Government Architecture (AGA)	Digital Transformation Agency
New Zealand	Government Enterprise Architecture for New Zealand (GEA-NZ)	Government Chief Digital Officer
Norway	National architecture principles for digitalisation of the public sector	Norwegian Digitalisation Agency (Digdir)
United States	Federal Enterprise Architecture Framework (FEAF)	Office of Management and Budget

The SAF works best when the NSI combines models at the level of elements, not documents:

- **Principles.** Map the SAF principles to the principles of the national reference model. Many will be equivalent. Where both exist, the NSI uses one set and records the mapping.
- **Capabilities and functions.** Map the SAF capability model to the capability or function model of the national framework, where one exists. This makes the NSI comparable with other government organisations as well as with other NSIs.
- **Standards.** Check which standards the national model prescribes, for example for exchange formats, identity or security. These standards become requirements or constraints in the NSI architecture.

When models conflict, a simple order of precedence helps:

1. Legislation comes first.
2. Mandatory government standards and reference models come next.
3. Within the statistical domain, the SAF and the statistical standards apply.
4. Generic frameworks and practices fill the remaining gaps.

A deviation from a higher level is a decision for the NSI's architecture governance, and is documented.

Toolkit: SAF-TK-G03 Reference model mapping

5. Legislation and regulation

Official statistics operate under strict legal frameworks. Legislation shapes what an NSI may collect, how it must protect data, and how it must publish. The SAF does not describe legislation, because it differs per country. It does describe where legislation enters the architecture.

Legislation enters the architecture through the motivation elements:

A legal provision that…	Is recorded as…	Example
Creates pressure to change	Driver	A new regulation that requires access to privately held data for statistics
Sets a limit on the design	Constraint	Statistical microdata may only be processed within the NSI's controlled environment
Imposes an obligation on a solution	Requirement	Personal data in a data acquisition process is minimised to what is needed
Expresses a lasting rule for all decisions	Principle	Privacy by design (SAF-P11)

This approach makes legislation traceable. From a legal provision, the NSI can follow the path to the requirements and constraints it creates, to the solutions that meet them, and to the evidence that shows compliance. In the other direction, every design decision that is driven by law can be traced back to its legal basis.

Three levels of legislation are relevant for most NSIs:

Level	Examples
International statistical frameworks	UN Fundamental Principles of Official Statistics; European Statistics Code of Practice; Regulation (EC) No 223/2009 on European statistics
Horizontal EU legislation	GDPR; Data Governance Act; Data Act; AI Act; NIS2 Directive; Interoperable Europe Act (Regulation (EU) 2024/903)
National legislation	National statistics act; archives legislation; national rules on government information security and open government

For NSIs outside the EU, the horizontal level consists of the corresponding national or regional legislation.

Compliance is not a separate layer. It is built into the architecture: legal requirements are part of the quality gates of the steady states, of the principles, and of the governance in value stream VS-07 *Governance and Compliance*. A legal register helps to keep the overview: a list of applicable legal provisions, with the drivers, constraints, requirements and principles they lead to.

Toolkit: SAF-TK-G04 Legal register

6. Conventions used in this document

Convention	Meaning
SAF-ST, SAF-D, SAF-G, SAF-O, SAF-P, SAF-M, SAF-V	Identifiers of reference elements in the Motivation layer
1.2.1, 2.1.2, …	Capability numbers in the capability model
VS-01 to VS-10, VS-01.1, …	Value streams and their stages
#0 to #6	Steady states of data and their boundary positions
SAF-A, SAF-C	Assumptions and constraints of the SAF itself (see *Constraints & Assumptions*)
Illustrative	Examples that show the form of an element. They are not requirements
PLACEHOLDER	Content that will be added, such as an ArchiMate view or a figure
Toolkit (planned)	A toolkit instrument that does not exist yet; it is listed in *Planned instruments* in the Toolkit
Toolkit	The SAF toolkit, a separate set of templates and instruments for applying the framework

Terms are defined in the SAF glossary, which takes precedence over definitions in this document (SAF-A4). The SAF uses British English spelling.